Fan-out benchmark - real simplicio-prompt kernel on sistema-sindico

Date: 2026-05-28   Model: meta-llama/llama-3.1-8b-instruct   temperature: 0.7   Task: password_strength

Engine: kernel.subagent_runtime.SubagentRuntime (PyPI simplicio-prompt 1.7.0). For each N, the kernel launches N real parallel LLM calls through LaneWorkerPool on the same prompt; each output is scored by real PHPUnit; the majority-vote outcome (sha256-mode of normalized code) is re-scored. Question: does the sp-default 64 help vs single? does 200 help vs 64?

Headline

N	Per-attempt	Uniq	Majority	Tokens	Cost	Elapsed
1	1/1 (100%)	1	PASS (1/1)	991	$0.0001	26.1s
8	8/8 (100%)	6	PASS (3/8)	8,008	$0.0005	28.7s
32	32/32 (100%)	16	PASS (14/32)	32,165	$0.0019	30.4s
64	64/64 (100%)	12	PASS (51/64)	65,205	$0.0039	24.6s
200	200/200 (100%)	24	PASS (156/200)	203,529	$0.0122	34.3s

Interpretation

Per-attempt pass = noise floor at temp=0.7 (single-call equivalent). Unique outputs = real diversity at this temperature (if all equal, fan-out adds nothing). Majority-vote = does picking the most frequent answer recover correctness? Cost / elapsed scale linearly with N for tokens; wall-clock grows slower because calls run in parallel.